Lesson 18: Compactness and the Baire Category Theorem

AI/ML Mathematics

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1 Motivation

Compactness is the precise generalization of finiteness to the topological setting. In finite-dimensional spaces, the Heine–Borel theorem tells us that closed and bounded sets are compact, and this compactness drives essentially every existence theorem: the extreme value theorem, the existence of minimizers, the existence of convergent subsequences. In infinite-dimensional spaces—the natural home of value functions, policies, and feature maps—compactness is far more subtle. Closed and bounded sets need not be compact, and new criteria are required.

The Baire category theorem is a different kind of existence tool. Rather than producing convergent subsequences, it establishes that complete metric spaces are "large" in a precise topological sense: they cannot be expressed as countable unions of "negligible" sets. This seemingly abstract result underpins the three pillars of Banach space theory—the Uniform Boundedness Principle, the Open Mapping Theorem, and the Closed Graph Theorem—all of which are proved in the next lesson.

Why Compactness and Baire Matter for RL

1. **Existence of optimal policies:** On compact action spaces, the supremum in the Bellman equation is attained (it becomes a maximum), guaranteeing the existence of an optimal deterministic policy Puterman [2014].
2. **Value function approximation:** Riesz's theorem (compact unit ball $\Leftrightarrow$ finite dimension) explains why we must restrict to finite-dimensional parameterizations in practice.
3. **Compactness of policy classes:** The Arzela–Ascoli theorem provides the criterion (uniform boundedness + equicontinuity) for a family of Lipschitz value functions to have convergent subsequences.
4. **Uniform error bounds:** The Uniform Boundedness Principle, which follows from Baire, converts pointwise convergence of TD estimates into uniform bounds on approximation error.

Prerequisites. Metric spaces, normed spaces, Banach spaces, bounded linear operators, and dual spaces (Lessons 15–17).

Outline. Section 2 introduces compactness and sequential compactness in metric spaces. Section 3 develops total boundedness and proves the fundamental equivalence theorem. Section 4 derives the Heine–Borel theorem as a corollary. Section 5 proves Riesz's lemma and the characterization of finite-dimensional spaces via compact unit balls. Section 6 proves the Arzela–Ascoli theorem. Section 7 introduces compact operators. Section 8 proves the Baire category theorem and derives immediate consequences. Section 9 provides exercises.

2 Compactness in Metric Spaces

Definition 2.1 (Compact Set)**.** A subset K of a metric space (X, d) is *compact* if every open cover of K has a finite subcover: whenever $K \subseteq \bigcup_{\alpha \in I} G_\alpha$ with each G_α open in X, there exist finitely many indices $\alpha_1, \ldots, \alpha_n \in I$ with $K \subseteq G_{\alpha_1} \cup \cdots \cup G_{\alpha_n}$.

Definition 2.2 (Sequential Compactness)**.** A subset K of a metric space (X, d) is *sequentially compact* if every sequence in K has a subsequence that converges to a point in K.

Remark 2.3. In general topological spaces, compactness and sequential compactness are distinct notions. In metric spaces, they are equivalent—this is the content of Theorem 3.4 below. We will prove this equivalence through the intermediate notion of total boundedness.

Proposition 2.4 (Basic Properties of Compact Sets)**.** *Let (X, d) be a metric space.*

(i) Every compact set is closed and bounded.

(ii) A closed subset of a compact set is compact.

(iii) A continuous image of a compact set is compact.

Proof. **(i) Closed:** Let K be compact and let $x \in X \setminus K$. For each $k \in K$, let $r_k = d(x, k)/2 > 0$. The balls $B(k, r_k)$ cover K, so finitely many suffice: $K \subseteq B(k_1, r_{k_1}) \cup \cdots \cup B(k_n, r_{k_n})$. Let $r = \min(r_{k_1}, \ldots, r_{k_n}) > 0$. Then $B(x, r) \cap K = \emptyset$ (if $y \in B(x, r) \cap B(k_i, r_{k_i})$, then $d(x, k_i) \leq d(x, y) + d(y, k_i) < r + r_{k_i} \leq 2r_{k_i} = d(x, k_i)$, a contradiction). So $X \setminus K$ is open, hence K is closed.

Bounded: The balls $B(x_0, n)$ for $n \in \mathbb{N}$ cover K (for any fixed x_0), so finitely many suffice, giving $K \subseteq B(x_0, N)$ for some N.

(ii) Let $F \subseteq K$ be closed and let $\{G_\alpha\}$ be an open cover of F. Then $\{G_\alpha\} \cup \{X \setminus F\}$ is an open cover of K. Extract a finite subcover, then remove $X \setminus F$ if present.

(iii) Let $f\colon X \to Y$ be continuous and $K \subseteq X$ compact. If $\{G_\alpha\}$ covers $f(K)$, then $\{f^{-1}(G_\alpha)\}$ is an open cover of K. Extract a finite subcover $f^{-1}(G_{\alpha_1}), \ldots, f^{-1}(G_{\alpha_n})$. Then $G_{\alpha_1}, \ldots, G_{\alpha_n}$ covers $f(K)$. □

3 Total Boundedness and the Compactness Equivalence

Definition 3.1 (Totally Bounded)**.** A subset A of a metric space (X, d) is *totally bounded* if for every $\varepsilon > 0$, there exist finitely many points $x_1, \ldots, x_n \in X$ such that $A \subseteq \bigcup_{i=1}^n B(x_i, \varepsilon)$. The set $\{x_1, \ldots, x_n\}$ is called a *finite ε-net* for A.

Proposition 3.2 (Totally Bounded Implies Bounded)**.** *Every totally bounded set is bounded. The converse fails in general.*

Proof. Suppose A is totally bounded. Choose a finite 1-net $\{x_1, \ldots, x_n\}$, so $A \subseteq \bigcup_{i=1}^n B(x_i, 1)$. For any $a, b \in A$, there exist i, j with $a \in B(x_i, 1)$ and $b \in B(x_j, 1)$, giving

$$d(a, b) \leq d(a, x_i) + d(x_i, x_j) + d(x_j, b) < 1 + \max_{i,j} d(x_i, x_j) + 1.$$

Hence $\operatorname{diam}(A) < \infty$.

For the failure of the converse, consider the closed unit ball $\overline{B} = \{x \in \ell^2 : \|x\| \leq 1\}$. This set is bounded with $\operatorname{diam}(\overline{B}) = 2$, but it is not totally bounded: the standard basis vectors e_n satisfy $\|e_n - e_m\| = \sqrt{2}$ for $n \neq m$, so no finite collection of balls of radius $\sqrt{2}/2$ can cover them all. □

Lemma 3.3 (Subsequence Extraction via Total Boundedness)**.** *If A is totally bounded and (x_n) is a sequence in A, then for every $\varepsilon > 0$, there exists a subsequence of (x_n) contained in a single ball of radius ε.*

Proof. Cover A by finitely many balls $B(y_1, \varepsilon), \ldots, B(y_m, \varepsilon)$. Since the sequence has infinitely many terms and the cover is finite, by the pigeonhole principle at least one ball contains infinitely many terms of the sequence. These terms form the desired subsequence. □

Theorem 3.4 (Characterization of Compactness in Metric Spaces)**.** *For a subset K of a metric space (X, d), the following are equivalent:*

(i) K is compact (every open cover has a finite subcover).

(ii) K is sequentially compact (every sequence has a convergent subsequence with limit in K).

(iii) K is complete and totally bounded.

Compactness Equivalence Theorem

In metric spaces, compactness, sequential compactness, and completeness plus total boundedness are all equivalent. This trichotomy is fundamental: compactness (open-cover) is used for abstract existence arguments, sequential compactness is used for extracting convergent subsequences, and completeness plus total boundedness is the criterion most easily verified in practice.

Proof. **(i) ⇒ (ii):** Let (x_n) be a sequence in K. Suppose for contradiction that no subsequence converges to a point in K. Then every $y \in K$ has a neighborhood $B(y, r_y)$ containing only finitely many terms of (x_n) (otherwise we could extract a subsequence converging to y). The collection $\{B(y, r_y) : y \in K\}$ is an open cover of K. By compactness, it has a finite subcover. But a finite union of sets, each containing finitely many terms, contains only finitely many terms total—contradicting the fact that (x_n) has infinitely many terms in K.

(ii) ⇒ (iii): *Completeness:* Let (x_n) be a Cauchy sequence in K. By sequential compactness, there is a subsequence $x_{n_k} \to x \in K$. A Cauchy sequence with a convergent subsequence converges (for any $\varepsilon > 0$, choose N so that $d(x_n, x_m) < \varepsilon/2$ for $n, m \geq N$, and choose k with $n_k \geq N$ and $d(x_{n_k}, x) < \varepsilon/2$; then $d(x_n, x) \leq d(x_n, x_{n_k}) + d(x_{n_k}, x) < \varepsilon$ for $n \geq N$). Hence $x_n \to x \in K$.

Total boundedness: Suppose K is not totally bounded. Then there exists $\varepsilon > 0$ such that no finite collection of ε-balls covers K. Pick $x_1 \in K$ arbitrarily. Having chosen $x_1, \ldots, x_n$, since $K \not\subseteq \bigcup_{i=1}^n B(x_i, \varepsilon)$, choose $x_{n+1} \in K \setminus \bigcup_{i=1}^n B(x_i, \varepsilon)$. Then $d(x_{n+1}, x_i) \geq \varepsilon$ for all $i \leq n$. The sequence (x_n) satisfies $d(x_n, x_m) \geq \varepsilon$ for all $n \neq m$, so no subsequence can be Cauchy (let alone convergent). This contradicts sequential compactness.

(iii) ⇒ (i): We prove this in two steps.

Step 1: Complete + totally bounded implies sequentially compact. Let (x_n) be a sequence in K. By Lemma 3.3 applied with $\varepsilon = 1$, extract a subsequence $(x_n^{(1)})$ lying in a ball of radius 1. Apply the lemma again to this subsequence with $\varepsilon = 1/2$ to get $(x_n^{(2)}) \subseteq B(y_2, 1/2)$ for some y_2. At stage k, extract $(x_n^{(k)}) \subseteq B(y_k, 1/k)$, a subsequence of $(x_n^{(k-1)})$. The diagonal subsequence $z_k = x_k^{(k)}$ is Cauchy: for $k, m \geq N$, both z_k and z_m belong to the ball $B(y_N, 1/N)$ (since z_k for $k \geq N$ is a term of $(x_n^{(N)})$), so $d(z_k, z_m) \leq 2/N \to 0$. By completeness of K, (z_k) converges to some $z \in K$.

Step 2: Sequentially compact implies compact (via the Lebesgue number lemma).

Claim (Lebesgue number): If K is sequentially compact and $\{G_\alpha\}$ is an open cover of K, then there exists $\delta > 0$ (the *Lebesgue number*) such that for every $x \in K$, $B(x, \delta) \subseteq G_\alpha$ for some α.

Proof of claim: Suppose not. Then for each $n \in \mathbb{N}$, there exists $x_n \in K$ with $B(x_n, 1/n) \not\subseteq G_\alpha$ for any α. By sequential compactness, extract a convergent subsequence $x_{n_k} \to x \in K$. Since $\{G_\alpha\}$ covers K, $x \in G_\beta$ for some β. Since G_β is open, $B(x, r) \subseteq G_\beta$ for some $r > 0$. For large k, $d(x_{n_k}, x) < r/2$ and $1/n_k < r/2$, so $B(x_{n_k}, 1/n_k) \subseteq B(x, r) \subseteq G_\beta$, contradicting the choice of x_{n_k}.

Now, given the Lebesgue number $\delta > 0$, cover K by finitely many balls of radius $\delta/2$ using total boundedness (which we established in Step 1). Each such ball has diameter at most δ, so it is contained in some G_α. The corresponding finitely many G_α's form a finite subcover. □

4 The Heine–Borel Theorem

Corollary 4.1 (Heine–Borel Theorem for $\mathbb{R}^n$)**.** *A subset of $\mathbb{R}^n$ is compact if and only if it is closed and bounded.*

Proof. (⇒) By Proposition 2.4(i), every compact set is closed and bounded.

($\Leftarrow$) Let $K \subseteq \mathbb{R}^n$ be closed and bounded. We verify the two conditions of Theorem 3.4(iii).

Complete: $\mathbb{R}^n$ is complete (every Cauchy sequence in $\mathbb{R}^n$ converges). A closed subset of a complete metric space is complete: if (x_n) is Cauchy in K, then $x_n \to x \in \mathbb{R}^n$ by completeness of $\mathbb{R}^n$, and $x \in K$ since K is closed.

Totally bounded: Since K is bounded, $K \subseteq [-M, M]^n$ for some $M > 0$. Given $\varepsilon > 0$, partition $[-M, M]^n$ into cubes of side length $\varepsilon/\sqrt{n}$. Each cube has diameter at most ε, so it is contained in a ball of radius ε centered at any of its points. Since the partition is finite, K is totally bounded.

By Theorem 3.4, K is compact. □

Heine–Borel and Finite-State RL

In finite-state RL with $|\mathcal{S}| = n$, the space of value functions is $\mathbb{R}^n$. The set of value functions bounded by $R_{\max}/(1-\gamma)$ is closed and bounded in $\mathbb{R}^n$, hence compact by Heine–Borel. This compactness guarantees that every bounded sequence of value function iterates has a convergent subsequence—a starting point for convergence analysis even before invoking contraction arguments.

5 Riesz's Lemma and Compactness in Infinite Dimensions

The Heine–Borel theorem fails spectacularly in infinite-dimensional normed spaces. The following lemma and its consequence explain exactly why.

Lemma 5.1 (Riesz's Lemma)**.** *Let Y be a* proper *closed subspace of a normed space X (i.e., $Y \subsetneq X$ and Y is closed). For every $\theta \in (0,1)$, there exists $x_\theta \in X$ with $\|x_\theta\| = 1$ and*

$$\operatorname{dist}(x_\theta, Y) = \inf_{y \in Y} \|x_\theta - y\| \geq \theta.$$

Proof. Since $Y \neq X$, there exists $z \in X \setminus Y$. Since Y is closed, $d = \operatorname{dist}(z, Y) = \inf_{y \in Y} \|z - y\| > 0$. Choose $y_0 \in Y$ with $\|z - y_0\| < d/\theta$ (possible since $d/\theta > d = \inf$ and the infimum is not necessarily attained). Define

$$x_\theta = \frac{z - y_0}{\|z - y_0\|}.$$

Then $\|x_\theta\| = 1$. For any $y \in Y$:

$$\|x_\theta - y\| = \frac{\|z - y_0 - \|z - y_0\| \cdot y\|}{\|z - y_0\|} = \frac{\|z - \underbrace{(y_0 + \|z - y_0\| \cdot y)}_{\in Y}\|}{\|z - y_0\|} \geq \frac{d}{\|z - y_0\|} > \frac{d}{d/\theta} = \theta.$$

Taking the infimum over $y \in Y$ gives $\operatorname{dist}(x_\theta, Y) \geq \theta$. □

Remark 5.2. One cannot always achieve $\theta = 1$ in Riesz's lemma. However, in reflexive spaces (in particular, in Hilbert spaces), the infimum $\operatorname{dist}(z, Y)$ is attained by the orthogonal projection, so one can take $\theta = 1$.

Theorem 5.3 (Compact Unit Ball Characterizes Finite Dimension (Riesz))**.** *The closed unit ball $\overline{B}(0,1) = \{x \in X : \|x\| \leq 1\}$ in a normed space X is compact if and only if $\dim X < \infty$.*

Proof. ($\Leftarrow$) If $\dim X = n < \infty$, then X is isomorphic to $\mathbb{R}^n$ as a normed space (all norms on a finite-dimensional space are equivalent). The closed unit ball is closed and bounded in $\mathbb{R}^n$, hence compact by the Heine–Borel theorem (Corollary 4.1).

($\Rightarrow$) We prove the contrapositive: if $\dim X = \infty$, then $\overline{B}(0,1)$ is not compact. We construct a sequence in $\overline{B}(0,1)$ with no convergent subsequence.

Pick $x_1 \in X$ with $\|x_1\| = 1$. Let $Y_1 = \operatorname{span}\{x_1\}$. Since $\dim X = \infty$, Y_1 is a proper subspace; it is closed because finite-dimensional subspaces of normed spaces are closed. By Riesz's lemma (Lemma 5.1) with $\theta = 1/2$, there exists x_2 with $\|x_2\| = 1$ and $\|x_2 - y\| \geq 1/2$ for all $y \in Y_1$. In particular, $\|x_2 - x_1\| \geq 1/2$.

Set $Y_2 = \operatorname{span}\{x_1, x_2\}$, again a proper closed subspace. By Riesz's lemma, there exists x_3 with $\|x_3\| = 1$ and $\operatorname{dist}(x_3, Y_2) \geq 1/2$. Continue inductively: at step n, set $Y_{n-1} = \operatorname{span}\{x_1, \ldots, x_{n-1}\}$ and find x_n with $\|x_n\| = 1$ and $\operatorname{dist}(x_n, Y_{n-1}) \geq 1/2$. Since $x_1, \ldots, x_{n-1} \in Y_{n-1}$, we have $\|x_n - x_m\| \geq 1/2$ for all $m < n$.

The sequence (x_n) lies in $\overline{B}(0,1)$, and any two terms are at least $1/2$ apart. No subsequence can be Cauchy, so no subsequence converges. Thus $\overline{B}(0,1)$ is not sequentially compact, hence not compact. □

Riesz's Theorem: Finite Dimension via Compactness

A normed space is finite-dimensional if and only if its closed unit ball is compact. This is one of the most fundamental structural results in functional analysis. It explains why bounded sets in infinite-dimensional spaces (such as the space of all value functions on an infinite state space) are never compact in the norm topology, and why alternative compactness criteria are essential.

Why We Need Function Approximation

The space of all bounded value functions $V : \mathcal{S} \to \mathbb{R}$ on an infinite state space, equipped with the sup norm, is infinite-dimensional. By Riesz's theorem, the closed ball $\{V : \|V\|_\infty \leq R_{\max}/(1-\gamma)\}$ is not compact. Consequently, a bounded sequence of value function iterates need not have a convergent subsequence.
This is the mathematical reason we restrict to finite-dimensional parameterizations (linear function approximation, neural networks): in a finite-dimensional subspace, bounded sets are compact, and convergent subsequences exist. The price we pay is approximation error (the "deadly triad" in TD learning involves precisely this tension).

6 The Arzela–Ascoli Theorem

Riesz's theorem tells us that norm-compactness fails in infinite dimensions. The Arzela–Ascoli theorem provides a practical *substitute*: under suitable regularity conditions, families of functions are relatively compact even in infinite-dimensional function spaces.

Definition 6.1 (Equicontinuity)**.** A family $\mathcal{F}$ of functions from a metric space (M, d_M) to $(\mathbb{R}, |\cdot|)$ is *equicontinuous at* $x_0 \in M$ if for every $\varepsilon > 0$, there exists $\delta > 0$ such that $|f(x) - f(x_0)| < \varepsilon$ for all $f \in \mathcal{F}$ and all $x \in M$ with $d_M(x, x_0) < \delta$. The family is *equicontinuous* (on M) if it is equicontinuous at every point of M.

When M is compact, equicontinuity implies *uniform equicontinuity*: for every $\varepsilon > 0$, there exists $\delta > 0$ such that $|f(s) - f(t)| < \varepsilon$ for all $f \in \mathcal{F}$ and all $s, t \in M$ with $d_M(s,t) < \delta$.

Definition 6.2 (Uniformly Bounded Family)**.** A family $\mathcal{F} \subseteq C(M)$ is *uniformly bounded* if $\sup_{f \in \mathcal{F}} \|f\|_\infty < \infty$.

Theorem 6.3 (Arzela–Ascoli Theorem)**.** *A subset $\mathcal{F} \subseteq (C[a,b], \|\cdot\|_\infty)$ is relatively compact (i.e., its closure $\operatorname{cl}(\mathcal{F})$ is compact) if and only if $\mathcal{F}$ is uniformly bounded and equicontinuous.*

Proof. **($\Leftarrow$) Uniformly bounded + equicontinuous $\Rightarrow$ relatively compact.**

By Theorem 3.4, it suffices to show that every sequence (f_n) in $\mathcal{F}$ has a uniformly convergent subsequence (the closure $\mathrm{cl}(\mathcal{F})$ inherits completeness from the Banach space $(C[a,b], \|\cdot\|_\infty)$, so sequential compactness of $\mathrm{cl}(\mathcal{F})$ will follow).

Step 1: Pointwise convergence on a countable dense set. Let $D = \{r_1, r_2, \ldots\}$ be a countable dense subset of $[a,b]$ (for instance, the rationals in $[a,b]$). The sequence $(f_n(r_1))_{n\geq 1}$ is bounded (by uniform boundedness of $\mathcal{F}$), so by the Bolzano–Weierstrass theorem it has a convergent subsequence $(f_{n_k^{(1)}}(r_1))_{k\geq 1}$. From this subsequence, extract a further subsequence convergent at r_2, obtaining $(f_{n_k^{(2)}})_{k\geq 1}$. Continue: at stage j, extract a subsequence $(f_{n_k^{(j)}})_{k\geq 1}$ of $(f_{n_k^{(j-1)}})_{k\geq 1}$ that converges at r_j.

Define the diagonal subsequence $g_k = f_{n_k^{(k)}}$. For each fixed $j \in \mathbb{N}$, the sequence $(g_k)_{k\geq j}$ is a subsequence of $(f_{n_k^{(j)}})_{k\geq 1}$, so $(g_k(r_j))_{k\geq 1}$ converges. Thus (g_k) converges pointwise on all of D.

Step 2: Upgrade to uniform convergence. Given $\varepsilon > 0$, by equicontinuity on the compact set $[a,b]$, choose $\delta > 0$ such that $|f(s) - f(t)| < \varepsilon/3$ for all $f \in \mathcal{F}$ and all $s, t \in [a,b]$ with $|s-t| < \delta$. Cover $[a,b]$ by finitely many intervals of length $< \delta$, and pick a point $r_{j_i} \in D$ in each interval, giving $r_{j_1}, \ldots, r_{j_m}$.

Since (g_k) converges at each r_{j_i}, choose K such that $|g_k(r_{j_i}) - g_l(r_{j_i})| < \varepsilon/3$ for all $k, l \geq K$ and all $i = 1, \ldots, m$.

For any $t \in [a,b]$, choose r_{j_i} with $|t - r_{j_i}| < \delta$. Then for $k, l \geq K$:

$$\begin{aligned}|g_k(t) - g_l(t)| &\leq |g_k(t) - g_k(r_{j_i})| + |g_k(r_{j_i}) - g_l(r_{j_i})| + |g_l(r_{j_i}) - g_l(t)| \\ &< \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon.\end{aligned}$$

Since t was arbitrary, $\|g_k - g_l\|_\infty < \varepsilon$ for all $k, l \geq K$. Hence (g_k) is Cauchy in $(C[a,b], \|\cdot\|_\infty)$ and therefore converges.

($\Rightarrow$) Relatively compact $\Rightarrow$ uniformly bounded + equicontinuous.

Uniformly bounded: If $\mathrm{cl}(\mathcal{F})$ is compact, it is bounded in the sup norm. Hence $\sup_{f\in\mathcal{F}} \|f\|_\infty \leq \sup_{g\in\mathrm{cl}(\mathcal{F})} \|g\|_\infty < \infty$.

Equicontinuous: Given $\varepsilon > 0$, cover $\mathrm{cl}(\mathcal{F})$ by finitely many open balls $B(g_1, \varepsilon/3), \ldots, B(g_N, \varepsilon/3)$ (possible since $\mathrm{cl}(\mathcal{F})$ is compact, hence totally bounded). Each $g_i \in C[a,b]$ is uniformly continuous on $[a,b]$, so choose $\delta_i > 0$ with $|g_i(s) - g_i(t)| < \varepsilon/3$ whenever $|s-t| < \delta_i$. Let $\delta = \min(\delta_1, \ldots, \delta_N) > 0$.

For any $f \in \mathcal{F}$, choose g_i with $\|f - g_i\|_\infty < \varepsilon/3$. For $|s-t| < \delta$:

$$|f(s) - f(t)| \leq |f(s) - g_i(s)| + |g_i(s) - g_i(t)| + |g_i(t) - f(t)| < \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon. \qquad \square$$

Arzela–Ascoli in Policy Optimization

In continuous-state RL, when we optimize over a class of Lipschitz value functions $\{V : |V(s) - V(t)| \leq L \cdot d(s,t)\}$ with a uniform Lipschitz constant L and uniform bound $\|V\|_\infty \leq C$, the Arzela–Ascoli theorem guarantees that any sequence from this class has a uniformly convergent subsequence. This is used in:

- **Fitted value iteration:** Ensuring that the sequence of fitted value functions has a convergent subsequence.
- **Actor-critic methods:** Guaranteeing compactness of policy classes when policies are parameterized by Lipschitz functions.
- **Regularization:** Adding Lipschitz constraints to neural network policies is precisely an application of Arzela–Ascoli to ensure compactness of the hypothesis class.

7 Compact Operators

Compact operators are the infinite-dimensional analogues of matrices. They map bounded sets to relatively compact sets, compensating for the failure of norm-compactness in infinite dimensions.

Definition 7.1 (Compact Operator)**.** A linear operator $T\colon X \to Y$ between normed spaces is *compact* if for every bounded subset $B \subseteq X$, the image $T(B)$ is relatively compact in Y (i.e., $\mathrm{cl}(T(B))$ is compact). Equivalently, T is compact if every bounded sequence (x_n) in X has a subsequence (x_{n_k}) such that (Tx_{n_k}) converges in Y.

Proposition 7.2 (Compact Operators are Bounded)**.** *Every compact operator is bounded (hence continuous).*

Proof. Suppose T is compact but not bounded. Then there exists a sequence (x_n) with $\|x_n\| = 1$ and $\|Tx_n\| \to \infty$. Since (x_n) is bounded, compactness of T provides a subsequence with (Tx_{n_k}) convergent, hence bounded. But $\|Tx_{n_k}\| \to \infty$, a contradiction. □

Proposition 7.3 (Finite-Rank Operators are Compact)**.** *If $T\colon X \to Y$ is a bounded linear operator with $\dim(\mathrm{ran}(T)) < \infty$, then T is compact.*

Proof. Let $B \subseteq X$ be bounded. Then $T(B)$ is a bounded subset of the finite-dimensional space $\mathrm{ran}(T)$. By the Heine–Borel theorem applied to $\mathrm{ran}(T) \cong \mathbb{R}^m$, $\mathrm{cl}(T(B))$ is compact. □

Example 7.4 (Integral Operators with Continuous Kernels)**.** Let $k\colon [a,b] \times [a,b] \to \mathbb{R}$ be continuous, and define $T\colon C[a,b] \to C[a,b]$ by

$$(Tf)(s) = \int_a^b k(s,t)\, f(t)\, dt.$$

Then T is compact. To verify this using Arzela–Ascoli: if (f_n) is bounded in $C[a,b]$ with $\|f_n\|_\infty \le M$, then:

- *Uniformly bounded:* $\|Tf_n\|_\infty \le (b-a) \cdot \|k\|_\infty \cdot M$.
- *Equicontinuous:* Since k is uniformly continuous on the compact set $[a,b]^2$, for any $\varepsilon > 0$ there exists $\delta > 0$ with $|k(s_1,t) - k(s_2,t)| < \varepsilon/((b-a)M)$ whenever $|s_1 - s_2| < \delta$. Then

$$|(Tf_n)(s_1) - (Tf_n)(s_2)| \le \int_a^b |k(s_1,t) - k(s_2,t)| \cdot |f_n(t)|\, dt < \varepsilon,$$

  uniformly in n.

By Arzela–Ascoli (Theorem 6.3), $\{Tf_n\}$ is relatively compact in $C[a,b]$.

Proposition 7.5 (Algebraic Properties of Compact Operators)**.** *Let X, Y, Z be normed spaces.*

(i) The set of compact operators from X to Y is a closed subspace of $\mathcal{B}(X,Y)$ (when Y is a Banach space).

(ii) If $T\colon X \to Y$ is compact and $S\colon Y \to Z$ is bounded, then ST is compact. Similarly, if S is compact and T is bounded, then ST is compact.

Proof. **(i)** Let (T_n) be a sequence of compact operators converging in operator norm to T. We show T is compact. Let (x_k) be a bounded sequence in X with $\|x_k\| \le M$. By a diagonal argument: since T_1 is compact, extract a subsequence $(x_k^{(1)})$ with $(T_1 x_k^{(1)})$ convergent. From this, extract $(x_k^{(2)})$ with $(T_2 x_k^{(2)})$ convergent. The diagonal sequence $y_k = x_k^{(k)}$ satisfies: $(T_n y_k)$ converges for each n.

We claim (Ty_k) is Cauchy. Given $\varepsilon > 0$, choose n with $\|T - T_n\| < \varepsilon/(3M)$. Choose K with $\|T_n y_k - T_n y_l\| < \varepsilon/3$ for $k, l \ge K$. Then

$$\|Ty_k - Ty_l\| \le \|Ty_k - T_n y_k\| + \|T_n y_k - T_n y_l\| + \|T_n y_l - Ty_l\| < \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon.$$

(ii) If (x_n) is bounded and T is compact, then (Tx_{n_k}) converges for some subsequence, so (STx_{n_k}) converges by continuity of S. The second claim is similar: if (x_n) is bounded, then (Tx_n) is bounded, and compactness of S extracts a convergent subsequence of (STx_n). □

8 The Baire Category Theorem

We now turn to a completely different tool. The Baire category theorem provides a way to prove that certain sets are "large" (or that certain properties hold "generically") using only completeness.

8.1 Nowhere Dense Sets and Baire Category

Definition 8.1 (Nowhere Dense Set)**.** A subset A of a metric space (X, d) is *nowhere dense* if $\operatorname{int}(\operatorname{cl}(A)) = \emptyset$: the closure of A has empty interior. Equivalently, A is nowhere dense if and only if for every non-empty open set $U \subseteq X$, there exists a non-empty open set $V \subseteq U$ with $V \cap A = \emptyset$.

Example 8.2 (Nowhere Dense Sets in $\mathbb{R}$)**.** (a) $\mathbb{Z}$ is nowhere dense in $\mathbb{R}$: $\operatorname{cl}(\mathbb{Z}) = \mathbb{Z}$, and $\mathbb{Z}$ contains no open interval.

(b) $\{1/n : n \in \mathbb{N}\}$ is nowhere dense: its closure is $\{0\} \cup \{1/n : n \in \mathbb{N}\}$, which contains no open interval.

(c) The Cantor set $\mathcal{C}$ is nowhere dense, uncountable, and closed (it equals its own closure, which has empty interior since every interval in $[0, 1]$ contains points not in $\mathcal{C}$).

(d) $\mathbb{Q}$ is *not* nowhere dense: $\operatorname{cl}(\mathbb{Q}) = \mathbb{R}$, and $\operatorname{int}(\mathbb{R}) = \mathbb{R} \neq \emptyset$.

Definition 8.3 (First Category (Meager) and Second Category)**.** A subset A of a metric space is of *first category* (or *meager*) if A can be written as a countable union of nowhere dense sets:

$$A = \bigcup_{n=1}^{\infty} A_n, \quad \text{each } A_n \text{ is nowhere dense.}$$

A set that is *not* of first category is said to be of *second category* (or *non-meager*, or *comeager* for the complement).

Remark 8.4. "First category" means topologically negligible ("small"). This is a fundamentally different notion from measure-theoretic negligibility (measure zero). The rationals $\mathbb{Q}$ are meager in $\mathbb{R}$ (as a countable union of singletons, each nowhere dense) and have Lebesgue measure zero. The Cantor set is nowhere dense (hence meager) but uncountable. There exist meager sets of full measure and comeager sets of measure zero.

8.2 Statement and Proof

Theorem 8.5 (Baire Category Theorem)**.** *Every complete metric space is of second category in itself. Equivalently:*

(a) If (X, d) is a complete metric space and $X = \bigcup_{n=1}^{\infty} F_n$ with each F_n closed, then at least one F_n has non-empty interior.

(b) If (X, d) is a complete metric space and $A_1, A_2, \ldots$ are nowhere dense, then $\bigcup_{n=1}^{\infty} A_n \neq X$.

Proof. We prove formulation (b). Suppose $A_1, A_2, \ldots$ are nowhere dense in a complete metric space (X, d). We construct a point $x \in X \setminus \bigcup_{n=1}^{\infty} A_n$.

Step 1. Since A_1 is nowhere dense, $\operatorname{cl}(A_1)$ has empty interior, so $X \setminus \operatorname{cl}(A_1)$ is a non-empty open set (if it were empty, then $\operatorname{cl}(A_1) = X$ and $\operatorname{int}(\operatorname{cl}(A_1)) = X \neq \emptyset$, contradicting nowhere density). Choose $x_1 \in X \setminus \operatorname{cl}(A_1)$ and $r_1 \in (0, 1)$ with $\overline{B}(x_1, r_1) \subseteq X \setminus \operatorname{cl}(A_1)$. In particular, $\overline{B}(x_1, r_1) \cap A_1 = \emptyset$.

Step 2. Since A_2 is nowhere dense and $B(x_1, r_1)$ is a non-empty open set, there exists a non-empty open ball $B(x_2, r_2)$ with $\overline{B}(x_2, r_2) \subseteq B(x_1, r_1)$ and $\overline{B}(x_2, r_2) \cap \operatorname{cl}(A_2) = \emptyset$, with $r_2 \in (0, 1/2)$.

Inductive step. At step n, find x_n and $r_n \in (0, 1/n)$ with

$$\overline{B}(x_n, r_n) \subseteq B(x_{n-1}, r_{n-1}) \quad \text{and} \quad \overline{B}(x_n, r_n) \cap A_n = \emptyset.$$

Step 3: The sequence (x_n) is Cauchy. For $m > n$, the nesting condition gives $x_m \in B(x_n, r_n)$, so $d(x_m, x_n) < r_n < 1/n \to 0$. Since X is complete, $x_n \to x$ for some $x \in X$.

Step 4: The limit avoids every A_n. For each fixed n, $x_m \in \overline{B}(x_n, r_n)$ for all $m \geq n$ (by nesting). Since $\overline{B}(x_n, r_n)$ is closed, $x = \lim_{m\to\infty} x_m \in \overline{B}(x_n, r_n)$. But $\overline{B}(x_n, r_n) \cap A_n = \emptyset$, so $x \notin A_n$.

Since this holds for every n, $x \notin \bigcup_{n=1}^{\infty} A_n$. Therefore $\bigcup_{n=1}^{\infty} A_n \neq X$. □

The Baire Category Theorem

A complete metric space cannot be written as a countable union of nowhere dense sets. Equivalently, if a complete metric space is expressed as a countable union of closed sets, at least one of them has non-empty interior. This theorem is the foundation of the three great theorems of Banach space theory: the Uniform Boundedness Principle, the Open Mapping Theorem, and the Closed Graph Theorem.

8.3 Consequences of the Baire Category Theorem

Corollary 8.6 (Countable Intersection of Dense Open Sets)**.** *In a complete metric space, the intersection of countably many dense open sets is dense (it is a dense G_δ set).*

Proof. Let $G_1, G_2, \ldots$ be dense open sets in a complete metric space X. We show that $G = \bigcap_{n=1}^{\infty} G_n$ is dense, i.e., G intersects every non-empty open set.

Let $U \subseteq X$ be non-empty and open. Since G_1 is dense, $U \cap G_1$ is non-empty and open. Choose x_1 and $r_1 \in (0, 1)$ with $\overline{B}(x_1, r_1) \subseteq U \cap G_1$. Since G_2 is dense, $B(x_1, r_1) \cap G_2$ is non-empty and open. Choose x_2 and $r_2 \in (0, 1/2)$ with $\overline{B}(x_2, r_2) \subseteq B(x_1, r_1) \cap G_2$. Continue inductively: at step n, choose x_n and $r_n \in (0, 1/n)$ with $\overline{B}(x_n, r_n) \subseteq B(x_{n-1}, r_{n-1}) \cap G_n$.

As in the proof of Theorem 8.5, (x_n) is Cauchy with limit $x \in \overline{B}(x_n, r_n) \subseteq G_n$ for each n, and $x \in \overline{B}(x_1, r_1) \subseteq U$. Hence $x \in U \cap \bigcap_{n=1}^{\infty} G_n$. □

Corollary 8.7 (The Rationals are Not a G_δ Set)**.** *The set $\mathbb{Q}$ of rationals cannot be expressed as a countable intersection of open subsets of $\mathbb{R}$.*

Proof. Suppose for contradiction that $\mathbb{Q} = \bigcap_{n=1}^{\infty} G_n$ with each G_n open. Since $\mathbb{Q}$ is dense, each G_n is dense. Also, for each $q \in \mathbb{Q}$, the set $\mathbb{R} \setminus \{q\}$ is open and dense. By Corollary 8.6, the intersection

$$\bigcap_{n=1}^{\infty} G_n \cap \bigcap_{q\in\mathbb{Q}} (\mathbb{R} \setminus \{q\}) = \mathbb{Q} \cap (\mathbb{R} \setminus \mathbb{Q}) = \emptyset$$

is dense in $\mathbb{R}$. But the empty set is not dense, a contradiction. □

8.4 Application Preview: The Uniform Boundedness Principle

Remark 8.8 (Preview: Uniform Boundedness Principle). The Baire category theorem yields the following powerful result (proved in the next lesson): if $\{T_\alpha\}_{\alpha\in I}$ is a family of bounded linear operators from a Banach space X to a normed space Y such that

$$\sup_{\alpha\in I} \|T_\alpha x\| < \infty \quad \text{for every } x \in X \quad \text{(pointwise boundedness)},$$

then $\sup_{\alpha\in I} \|T_\alpha\| < \infty$ (uniform boundedness). The proof writes $X = \bigcup_{n=1}^{\infty} \{x \in X : \sup_\alpha \|T_\alpha x\| \leq n\}$. Each set in the union is closed (by continuity of the T_α). By the Baire category theorem, at least one has non-empty interior, and this leads to the uniform bound.

Baire Category and Genericity in RL

The Baire category theorem is used in RL theory to prove "genericity" results—that certain properties hold for "most" instances in a topological sense. For example:

- One can show that for a *generic* MDP (in the space of transition kernels and reward functions), the optimal policy is unique Puterman [2014]. The set of MDPs with multiple optimal policies is meager.
- The Uniform Boundedness Principle (a consequence of Baire) is used in the analysis of temporal-difference learning to conclude that pointwise convergence of TD estimates implies uniform bounds on the approximation error.
- The classical result that "most" continuous functions are nowhere differentiable (Exercise 9.9) illustrates that generic behavior can be highly irregular—a cautionary tale for assuming smoothness of value functions.

9 Exercises

Exercise 9.1 (Compact Sets are Closed and Bounded)**.** Let (X, d) be a metric space and $K \subseteq X$ compact.

(a) Show directly from the open cover definition that K is closed.
(b) Show that K is bounded. Give an example of a closed and bounded set in an infinite-dimensional normed space that is not compact.

Exercise 9.2 (Total Boundedness in $\mathbb{R}^n$)**.** Show that a subset $A \subseteq \mathbb{R}^n$ is totally bounded if and only if it is bounded. (This equivalence fails in infinite-dimensional spaces.)

Exercise 9.3 (Sequential Compactness of $[0,1]^{\mathbb{N}}$)**.** The space $[0,1]^{\mathbb{N}}$ of sequences with values in $[0,1]$ can be equipped with the metric $d(x,y) = \sum_{n=1}^{\infty} 2^{-n}|x_n - y_n|$.

(a) Show that $([0,1]^{\mathbb{N}}, d)$ is a complete metric space.
(b) Use a diagonal argument to show that $[0,1]^{\mathbb{N}}$ is sequentially compact (hence compact).
(c) Explain why this does not contradict Riesz's theorem.

Exercise 9.4 (Riesz's Lemma is Sharp in General)**.** Consider $X = (C[0,1], \|\cdot\|_\infty)$ and the closed subspace $Y = \{f \in C[0,1] : f(0) = 0\}$.

(a) Show that Y is indeed a proper closed subspace of X.
(b) For $\theta \in (0,1)$, find an explicit $f_\theta \in X$ with $\|f_\theta\| = 1$ and $\operatorname{dist}(f_\theta, Y) \geq \theta$.
(c) Show that in ℓ^2, Riesz's lemma can be strengthened to $\theta = 1$ for any proper closed subspace.
Hint: Use the orthogonal projection onto Y.

Exercise 9.5 (Equicontinuity and Lipschitz Families)**.** Let $L > 0$ and $\mathcal{F}_L = \{f \in C[0,1] : |f(s) - f(t)| \leq L|s-t| \text{ for all } s,t, \text{ and } |f(0)| \leq 1\}$.

(a) Show that $\mathcal{F}_L$ is equicontinuous and uniformly bounded (with $\|f\|_\infty \leq 1 + L$).
(b) Conclude by Arzela–Ascoli that $\mathcal{F}_L$ is relatively compact in $(C[0,1], \|\cdot\|_\infty)$.
(c) Show that $\mathcal{F}_L$ is closed, hence compact.

Exercise 9.6 (The Bellman Operator and Compact Operators)**.** Let $\mathcal{S} = [0,1]$, $\mathcal{A} = \{a\}$ (single action), and suppose the transition kernel has a continuous density $p(\cdot|s) \in C[0,1]$ for each s, with $(s,s') \mapsto p(s'|s)$ jointly continuous. The Bellman operator for a fixed policy is

$$(TV)(s) = r(s) + \gamma \int_0^1 p(s'|s)\, V(s')\, ds',$$

where $r \in C[0,1]$ and $\gamma \in (0,1)$.

(a) Show that the linear part $L\colon C[0,1] \to C[0,1]$, $(LV)(s) = \gamma \int_0^1 p(s'|s)\, V(s')\, ds'$, is a compact operator.
(b) Show that T is a contraction on $(C[0,1], \|\cdot\|_\infty)$ with factor γ.
(c) Explain why compactness of L ensures that the sequence $V, TV, T^2V, \ldots$ has a uniformly convergent subsequence, even before invoking the contraction mapping theorem.

Exercise 9.7 (Baire Category: $\mathbb{R}$ is Uncountable)**.** Use the Baire category theorem to prove that $\mathbb{R}$ is uncountable. (Write $\mathbb{R} = \bigcup_n \{r_n\}$ if $\mathbb{R}$ were countable, and derive a contradiction.)

Exercise 9.8 (Baire Category Application)**.** Use the Baire category theorem to show that $\mathbb{R}$ cannot be written as a countable union of sets, each of which is closed and has empty interior. Give an example showing that $\mathbb{R}$ *can* be written as a countable union of closed sets (some of which necessarily have non-empty interior).

Exercise 9.9 (Nowhere Differentiable Functions are Generic)**.** This exercise outlines the classical Baire category proof that "most" continuous functions are nowhere differentiable.

For $n \in \mathbb{N}$, let

$$A_n = \big\{ f \in C[0,1] : \text{there exists } x \in [0,1] \text{ with } |f(x+h) - f(x)| \le n|h| \text{ for all small } |h| \big\}.$$

(a) Show that A_n is a closed subset of $(C[0,1], \|\cdot\|_\infty)$.
(b) Show that A_n has empty interior.
Hint: Approximate any $f \in A_n$ uniformly by a piecewise-linear function with slopes exceeding n.
(c) Conclude that $\bigcup_{n=1}^\infty A_n$ is meager, so the set of functions that are somewhere differentiable is meager. The "generic" continuous function is nowhere differentiable.

Exercise 9.10 (Compact Operators Form a Closed Ideal)**.** Let X be a Banach space.
(a) Show that if $T\colon X \to X$ is compact and $S\colon X \to X$ is bounded, then both TS and ST are compact.
(b) Show that the set $\mathcal{K}(X)$ of compact operators on X is a closed two-sided ideal in $\mathcal{B}(X)$.
(c) Give an example of a bounded operator that is *not* compact.

10 Summary of Key Results

Result	**Statement (Informal)**	**RL/ML Relevance**
Compactness equivalence (Thm. 3.4)	Compact $\Leftrightarrow$ seq. compact $\Leftrightarrow$ complete + totally bounded	Existence of optimal policies on compact spaces
Heine–Borel (Cor. 4.1)	In $\mathbb{R}^n$: compact $\Leftrightarrow$ closed + bounded	Finite-state RL convergence
Riesz's theorem (Thm. 5.3)	Compact unit ball $\Leftrightarrow$ finite-dimensional	Necessity of function approximation
Arzela–Ascoli (Thm. 6.3)	Relatively compact $\Leftrightarrow$ uniformly bounded + equicontinuous	Compactness of Lipschitz policy classes
Compact operators (Def. 7.1)	Map bounded sets to relatively compact sets	Bellman operator with continuous kernel
Baire category (Thm. 8.5)	Complete metric space $\neq$ countable union of nowhere dense sets	Underpins Uniform Boundedness Principle

Looking Ahead

With compactness, the Arzela–Ascoli theorem, and the Baire category theorem in hand, we are ready for:

- **Lesson 19:** Inner product spaces and Hilbert spaces—the geometric structure needed for projections, least-squares, and RKHS theory.
- **Lesson 20:** The Projection Theorem and Riesz Representation—the tools for orthogonal projection in value function approximation.
- **Lesson 21:** The fundamental theorems of Banach space theory (Hahn–Banach, Uniform Boundedness, Open Mapping, Closed Graph), all powered by the Baire category theorem.

References

Martin L. Puterman. *Markov Decision Processes: Discrete Stochastic Dynamic Programming*. Wiley, 2014.